Maulik C. Madhavi

Video Analytics Researcher, Lead Product Engineer, NCS Pte Ltd, Singapore
Email: maulikmadhavi@gmail.com — Phone: +65-85443713
Website: https://sites.google.com/site/maulikcmadhavi/ — GitHub: https://github.com/maulikmadhavi

Research and Technical Experience

Video Analytics Researcher
NCS Pte Ltd, Singapore
April 2021 - Present

- Face indexing system to support million scale face search
- Development of vision-language applications supporting image captioning and vision question-answering
- Human person attribute recognition and audio classification for video applications

Research Fellow
National University of Singapore (NUS)
December 2017 - April 2021

- Chatbot for autonomous vehicle
- Text-dependent speaker verification
- ASR for healthcare calls
- Form filling system
- PI: Prof. Haizhou Li
- Mentored and co-supervised final year undergraduate students

Project Personnel
Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar
April 2012 - June 2014

- Prosodically Guided Phonetic Engine Development for Gujarati and Marathi languages
- PI: Prof. Hemant Patil

Education

Ph.D., Information and Communication Technology (ICT)
Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar
2011 - 2017 — CPI: 9.75/10
M.Tech, Information Communication Technology (ICT)
Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar
2009 - 2011 — CPI: 8.26/10
B.E., Electronics and Communication (EC)
V. V. P. Engineering College, Rajkot (Saurashtra University)
2005 - 2009 — Percentage: 69.73%
H.S.C.
Gujarat Secondary and Higher Secondary Education Board
2004 - 2005 — Percentage: 85.80%
S.S.C.
Gujarat Secondary and Higher Secondary Education Board
2002 - 2003 — Percentage: 82.14%

Skills

Research Areas: Speech signal processing, voice search, speaker verification, spoken language understanding, deep learning for spoken language processing, million-scale indexing

Programming Languages: Bash, C/C++, Python, MATLAB, Git, Docker, Kaldi, TensorFlow, PyTorch
Operating Systems: Windows and Linux

Research Publications

International Journals

1. M. C. Madhavi and H. A. Patil, "Vocal Tract Length Normalization using a Gaussian mixture model framework for query-by-example spoken term detection," *Computer, Speech and Language*, Elsevier, vol. 58, pp. 175-202, November 2019.
2. M. C. Madhavi and H. A. Patil, "Design of mixture of GMMs for query-by-example spoken term detection," *Computer, Speech and Language*, Elsevier, vol. 52, pp. 41-55, November 2018.
3. H. A. Patil and M. C. Madhavi, "Combining evidences from magnitude and phase information using VTEO for person recognition using humming," *Computer Speech and Language*, Elsevier, vol. 52, pp. 225-256, November 2018.
4. M. C. Madhavi and H. A. Patil, "Partial matching and search space reduction for QbE-STD," *Computer Speech & Language*, Elsevier, vol. 45, pp. 58-82, September 2017.
5. H. A. Patil, M. C. Madhavi, and K. K. Parhi, "Static and dynamic information derived from source and system features for person recognition from humming," *Int. Journal of Speech Technology*, Springer, vol. 15, no. 3, pp. 393-406, 2012.

Book Chapters

6. M. C. Madhavi, H. A. Patil, and N. Bhendawade, "Spoken Keyword Retrieval using Source and System Features," *PReMI, Lecture Notes in Computer Science (LNCS)*, pp. 333-341, 2017.
7. M. C. Madhavi, S. Sharma, and H. A. Patil, "VTLN Using different warping functions for template matching," *Machine Intelligence and Big Data in Industry*, Springer International Publishing, pp. 111-121, 2016.
8. M. C. Madhavi, S. Sharma, and H. A. Patil, "Vocal tract length normalization features for audio search," *Int. Conf. Text, Speech, and Dialogue, TSD*, pp. 387-395, 2015.
9. Y. Gaur, M. C. Madhavi, and H. A. Patil, "Speaker recognition using sparse representation via superimposed features," *PReMI, Lecture Notes in Computer Science (LNCS)*, vol. 8251, pp. 140-147, Springer-Verlag, 2013.
10. H. A. Patil, M. C. Madhavi, R. Jain, and A. Jain, "Combining evidences from temporal and spectral features for person recognition from humming," *PerMIn, Lecture Notes in Computer Science (LNCS)*, vol. 7143, pp. 321-328, Springer-Verlag, 2012.

International Conferences

11. B. Sharma, M. Madhavi, X. Zhou, and H. Li, "Exploring teacher-student learning approach for multi-lingual speech-to-intent classification," *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, December 2021, pp. 419-426.
12. Y. Jiang, B. Sharma, M. Madhavi, and H. Li, "Knowledge Distillation from BERT Transformer to Speech Transformer for Intent Classification," *Interspeech*, Brno, Czech Republic, August-September 2021, pp. 4713-4717.
13. R. Das, M. Madhavi, and H. Li, "Diagnosis of COVID-19 using Auditory Acoustic Cues," *Interspeech*, Brno, Czech Republic, August-September 2021, pp. 921-925.
14. X. Qian, M. Madhavi, Z. Pan, J. Wang, and H. Li, "Multi-target DoA estimation with an audio-visual fusion mechanism," *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Toronto, Canada, June 2021, pp. 4280-4284.
15. B. Sharma, M. Madhavi, and H. Li, "Leveraging Acoustic and Linguistic Embeddings from Pretrained Speech and Language Models for Intent Classification," *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Toronto, Canada, June 2021, pp. 7498-7502.
16. Y. Ong, M. Madhavi, and K. Chan, "OpenNLU: Open-Source Web-Interface NLU Toolkit for Development of Conversational Agent," *Asia-Pacific Signal and Information Processing Association (APSIPA) Annual Summit and Conference (ASC)*, Auckland, New Zealand, December 2020, pp. 381-385.
17. N. Shah, Sreeraj R, M. Madhavi, N. Shah, and H. Patil, "Query-by-Example Spoken Term Detection using Generative Adversarial Network," *APSIPA Annual Summit and Conference (ASC)*, Auckland, New Zealand, December 2020, pp. 644-648.

18. W. Lin, M. Madhavi, R. Das, and H. Li, "Transformer-based Arabic Dialect Identification," *International Conference on Asian Language Processing (IALP)*, Kuala Lumpur, Malaysia, December 2020, pp. 192-196.
19. T. Liu, R. Das, M. Madhavi, S. Shen, and H. Li, "Speaker-Utterance Dual Attention for Speaker and Utterance Verification," *Interspeech*, Shanghai, China, October 2020, pp. 4293-4297.
20. R. Sheelvant, B. Sharma, M. Madhavi, R. Das, S.R.M. Prasanna, and H. Li, "RSL2019: A Realistic Speech Localization Corpus," *Oriental COCOSDA*, Cebu City, Philippines, October 2019, pp. 1-6.
21. T. Liu, M. Madhavi, R. Das, and H. Li, "A Unified Framework for Speaker and Utterance Verification," *Interspeech*, Graz, Austria, September 2019, pp. 4320-4324.
22. M. Madhavi, T. Zhan, H. Li, and M. Yuan, "First Leap Towards Development of Dialogue System for Autonomous Bus," *International Workshop on Spoken Dialogue Systems Technology (IWSDS)*, Sicily, Italy, April 2019, pp. 1-6.
23. R. Das, M. C. Madhavi, and H. Li, "Compensating Utterance Information in Fixed Phrase Speaker Verification," *APSIPA Annual Summit and Conference (ASC)*, Honolulu, Hawaii, USA, November 2018, pp. 1708-1712.
24. P. A. Tapkir, M. R. Kamble, H. A. Patil, and M. C. Madhavi, "Replay Spoof Detection using Power Function Based Features," *APSIPA Annual Summit and Conference (ASC)*, Honolulu, Hawaii, USA, November 2018, pp. 1019-1023.
25. N. Shah, M. C. Madhavi, and H. Patil, "Unsupervised Vocal Tract Length Warped Posterior Features for Non-Parallel Voice Conversion," *Interspeech*, Hyderabad, India, September 2-6, 2018, pp. 1968-1972.
26. M. C. Madhavi and H. A. Patil, "VTLN-Warped Gaussian posteriorgram for QbE-STD," *25th European Signal Processing Conference (EUSIPCO)*, Kos Island, Greece, August 28-September 2, 2017, pp. 593-597.
27. M. C. Madhavi and H. A. Patil, "Two Stage Zero-resource Approaches for QbE-STD," *International Conference on Advances in Pattern Recognition (ICAPR)*, Bangalore, India, December 28-30, 2017.
28. M. C. Madhavi and H. A. Patil, "Combining Evidences from Detection Sources for Query-by-Example Spoken Term Detection," *APSIPA Annual Summit and Conference*, Kuala Lumpur, Malaysia, December 12-15, 2017.
29. A. Rajpal, T. B. Patel, H. B. Sailor, M. C. Madhavi, Hemant A. Patil, and Hiroya Fujisaki, "Native language identification using spectral and source-based features," *Interspeech*, San Francisco, USA, September 8-12, 2016, pp. 369-372.
30. M. C. Madhavi and H. A. Patil, "Modification in sequential dynamic time warping for fast computation of query-by-example spoken term detection task," *International Conference on Signal Processing and Communications (SPCOM)*, Bangalore, India, June 12-15, 2016, pp. 1-6.
31. M. C. Madhavi, H. A. Patil, and B. B. Vachhani, "Spectral transition measure for detection of obstruents," *23rd European Signal Processing Conference (EUSIPCO)*, Nice, France, August 31 - September 4, 2015, pp. 330-334.
32. H. B. Sailor, M. C. Madhavi, and H. A. Patil, "Significance of phase-based features for person recognition using humming," *2nd International Conference on Perception and Machine Intelligence (PerMin)*, Kolkata, February 26-27, 2015, pp. 99-103.
33. B. Vachhani, K. D. Malde, M. C. Madhavi, and H. A. Patil, "A spectral transition measure based MELCEPSTRAL features for obstruent detection," *International Conference on Asian Language Processing (IALP)*, Kuching, Malaysia, 2014, pp. 50-53.
34. S. Sharma, M. C. Madhavi, and H. A. Patil, "Vocal Tract Length Normalization for Vowel Recognition in Low Resource Languages," *International Conference on Asian Language Processing (IALP)*, Kuching, Malaysia, 2014, pp. 54-57.
35. M. C. Madhavi, S. Sharma, and H. A. Patil, "Development of language resources for speech application in Gujarati and Marathi," *International Conference on Asian Language Processing (IALP)*, Kuching, Malaysia, 2014, pp. 115-118.
36. A. Undhad, H. Patil, and M. C. Madhavi, "Exploiting speech source information for vowel landmark detection for low resource language," *9th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, Singapore, September 12-14, 2014, pp. 546-550.
37. N. Shah, H. Patil, M. Madhavi, H. Sailor, and T. Patel, "Deterministic annealing EM algorithm for developing TTS System in Gujarati," *9th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, Singapore, September 12-14, 2014, pp. 526-530.
38. M. Madhavi and H. A. Patil, "Exploiting variable length Teager energy operator in melcepstral features for person recognition from humming," *9th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, Singapore, September 12-14, 2014, pp. 624-628.
39. S. Sharma, M. C. Madhavi, and H. A. Patil, "Development of vocal tract length normalized phonetic engine for Gujarati and Marathi languages," *17th Oriental COCOSDA*, Phuket, Thailand, September 10-12, 2014.

40. K. D. Malde, B. B. Vachhani, M. C. Madhavi, N. H. Chhayani, and H. A. Patil, "Development of speech corpora in Gujarati and Marathi for phonetic transcription," *International Conference Oriental COCOSDA*, Gurgaon, India, 2013, pp. 1-6.
41. H. A. Patil, M. C. Madhavi, K. D. Malde, and B. B. Vachhani, "Phonetic transcription of Fricatives and Plosives for Gujarati and Marathi Languages," *International Conference on Asian Language Processing (IALP)*, Hanoi, Vietnam, November 13-15, 2012, pp. 177-180.
42. H. A. Patil, M. C. Madhavi, and N. H. Chhayani, "Person Recognition Using Humming, Singing and Speech," *International Conference on Asian Language Processing (IALP)*, Hanoi, Vietnam, November 13-15, 2012, pp. 149-152.
43. H. A. Patil and M. C. Madhavi, "Significance of magnitude and phase information via VTEO for humming based biometrics," *International Conference on Biometrics (ICB)*, New Delhi, India, 2012, pp. 372-377.
44. H. A. Patil, M. C. Madhavi, and K. K. Parhi, "Combining evidence from spectral and source-like features for person recognition from humming," *Interspeech*, Florence, Italy, August 27-31, 2011, pp. 369-372.

Awards and Achievements

- IAPR (International Association for Pattern Recognition) Travel Scholarship for presenting paper in International Conference on Biometrics, ICB 2012, Delhi, India. Paper: "Significance of Magnitude and Phase Information via VTEO for Humming Based Biometrics." (Grant reference: IAPR009)
- Graduate Aptitude Test in Engineering (GATE) 2009 qualified in Electronics and Communication Engineering with All India Rank 2142

Personal Information

Date of Birth: March 4, 1988
Current Address: Blk 565 Choa Chu Kang Street 52, #10-128, Singapore, 680565
Indian Residential Address: "AUM", 1-Wankaner Society, Jamanagar Road, Rajkot, Gujarat, India, 360006